

# Class 10 NCERT Maths

## CLASS 10 NCERT MATHS

Chapter 11: Areas Related to Circles

### CHAPTER 11: AREAS RELATED TO CIRCLES

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A simplified, detailed guide in Hinglish

For students who missed Class 7-8-9

Based on the official NCERT English Medium textbook

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## SECTION 1: FOUNDATION

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### 1.1 CIRCLE KE BASIC FORMULAS (YAAD RAKH)

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Circumference (perimeter) =  $2 \times \pi \times r$  (or  $\pi \times d$ )

Area of circle =  $\pi \times r^2$

Diameter =  $2r$

$r$  = radius

$\pi = 22/7$  (or 3.14)

### 1.2 $\pi$ (PI) KYA HAI?

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$\pi$  = circumference / diameter ka ratio.

Har circle ke liye SAME hota hai = 3.14159...

### EXAM MEIN:

- Radius 7 ka multiple ho -> use  $\pi = 22/7$
- Warna -> use  $\pi = 3.14$

### 1.3 EXAMPLE: BASIC

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EXAMPLE 1: Radius 7 cm circle ka area aur circumference

Area =  $\pi r^2 = (22/7) \times 7 \times 7 = 22 \times 7 = 154 \text{ cm}^2$

Circumference =  $2 \pi r = 2 \times (22/7) \times 7 = 44 \text{ cm}$

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## SECTION 2: SECTOR OF A CIRCLE

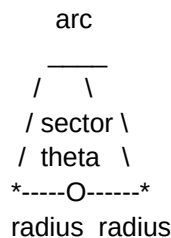
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### 2.1 SECTOR KYA HAI?

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SECTOR = circle ka ek "pizza slice" - do radii aur ek arc ke beech ka area.

Angle theta us slice ka angle hai (centre pe).



## 2.2 FORMULAS (theta in degrees)

Area of sector =  $(\theta/360) \times \pi \times r^2$   
Length of arc =  $(\theta/360) \times 2 \times \pi \times r$   
Perimeter of sector = arc length +  $2r$

LOGIC: Pura circle 360 degree hai. Sector us ka  $\theta/360$  hissa hai. So area aur arc bhi utna fraction.

## 2.3 EXAMPLES

EXAMPLE 2: Radius 21 cm, angle 60. Sector area.

$$\begin{aligned}\text{Area} &= (60/360) \times (22/7) \times 21 \times 21 \\ &= (1/6) \times (22/7) \times 441 \\ &= (1/6) \times 22 \times 63 \\ &= (1/6) \times 1386 \\ &= 231 \text{ cm}^2\end{aligned}$$

EXAMPLE 3: Radius 7 cm, angle 90. Arc length.

$$\begin{aligned}\text{Arc} &= (90/360) \times 2 \times (22/7) \times 7 \\ &= (1/4) \times 44 \\ &= 11 \text{ cm}\end{aligned}$$

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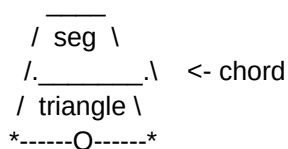
## SECTION 3: SEGMENT OF A CIRCLE

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### 3.1 SEGMENT KYA HAI?

SEGMENT = chord aur arc ke beech ka area (sector mein

se triangle nikaal do).



### 3.2 FORMULA

Area of (minor) segment  
= Area of sector - Area of triangle

Area of sector =  $(\theta/360) \times \pi \times r^2$   
Area of triangle =  $(1/2) \times r^2 \times \sin(\theta)$

So:  
Segment =  $(\theta/360) \pi r^2 - (1/2) r^2 \sin(\theta)$

### 3.3 EXAMPLE

EXAMPLE 4: Radius 10 cm, angle 90. Minor segment area.  
( $\pi = 3.14$ )

Sector area =  $(90/360) \times 3.14 \times 10 \times 10$   
=  $(1/4) \times 314 = 78.5 \text{ cm}^2$

Triangle area =  $(1/2) \times r^2 \times \sin 90$   
=  $(1/2) \times 100 \times 1 = 50 \text{ cm}^2$

Segment =  $78.5 - 50 = 28.5 \text{ cm}^2$

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## SECTION 4: MINOR VS MAJOR

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- MINOR sector/segment: chhota wala (angle < 180)
- MAJOR sector/segment: bada wala (angle > 180)
- Major area = total circle area - minor area
- Major angle = 360 - minor angle

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## Q AND A TIME

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Q1. Radius 14 cm circle ka area aur circumference.  
( $\pi = 22/7$ )

Q2. Radius 6 cm, angle 60. Sector area. ( $\pi = 3.14$ )

Q3. Radius 14 cm, angle 90. Arc length. ( $\pi = 22/7$ )

Q4. Sector ka perimeter kaise nikaalte hain?  
(formula bata)

Q5. Radius 4 cm, angle 90. Minor segment area.  
( $\pi = 3.14$ ,  $\sin 90 = 1$ )

Q6. Minor aur major sector mein kya difference hai?

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## SUMMARY

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1. Circumference =  $2\pi r$ , Area =  $\pi r^2$
2.  $\pi = 22/7$  (radius 7 ka multiple) ya 3.14
3. Sector area =  $(\theta/360) \times \pi r^2$
4. Arc length =  $(\theta/360) \times 2\pi r$
5. Perimeter of sector = arc length +  $2r$
6. Segment area = sector area - triangle area  
=  $(\theta/360) \pi r^2 - (1/2) r^2 \sin(\theta)$
7. Major = total - minor

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## TIPS FOR EXAM

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1.  $\theta/360$  ka fraction pehle simplify kar lo  
(jaise  $90/360 = 1/4$ ) - calculation easy.
2.  $\pi = 22/7$  vs 3.14 sahi choose karo - radius dekho.
3. Segment = sector - triangle. Triangle ke liye  
 $(1/2) r^2 \sin(\theta)$  formula yaad rakh.

4. Units: area =  $\text{cm}^2$ , length/perimeter = cm.

5. Combination figures (square mein circle, etc.)  
mein areas ko ADD/SUBTRACT karo - diagram banao.